

🔗 Practical 4

Aim: Write a python program to demonstrate various data visualization techniques.

```
# Import necessary libraries import pandas as pd import
matplotlib.pyplot as plt import seaborn as sns

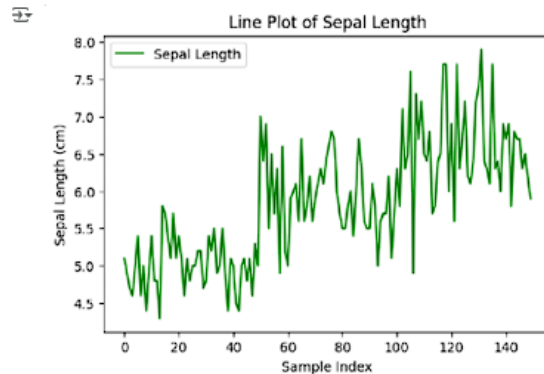
# Load a sample dataset df = sns.load_dataset('iris') # Built-in dataset with
150 flower samples

# Set seaborn style sns.set(style="whitegrid")
```

▲ 1. Line Plot

```
# Load dataset df = sns.load_dataset('iris')

# Line Plot plt.figure(figsize=(6, 4))
plt.plot(df['sepal_length'], label='Sepal Length', color='green')
plt.title('Line Plot of Sepal Length') plt.xlabel('Sample Index')
plt.ylabel('Sepal Length (cm)') plt.legend() plt.show()
```

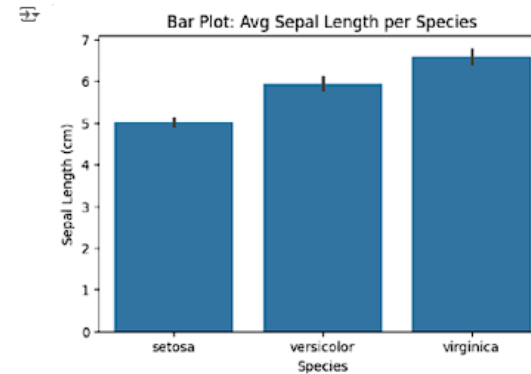


Double-click (or enter) to edit

▲ 2. Bar Plot

```
# Load dataset df = sns.load_dataset('iris')

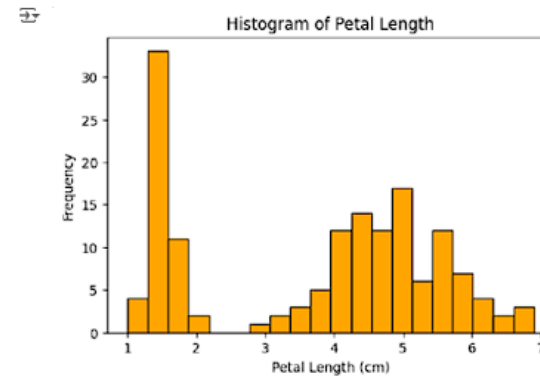
# Bar Plot plt.figure(figsize=(6, 4))
sns.barplot(x='species', y='sepal_length', data=df)
plt.title('Bar Plot: Avg Sepal Length per Species')
plt.xlabel('Species') plt.ylabel('Sepal Length (cm)')
plt.show()
```



▲ 3. Histogram

```
# Load dataset df = sns.load_dataset('iris')

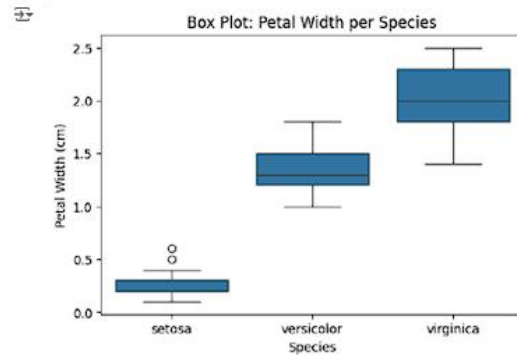
# Histogram plt.figure(figsize=(6, 4)) plt.hist(df['petal_length'], bins=20,
color='orange', edgecolor='black') plt.title('Histogram of Petal Length')
plt.xlabel('Petal Length (cm)') plt.ylabel('Frequency') plt.show()
```



▲ 4. Box Plot

```
# Load dataset df = sns.load_dataset('iris')
```

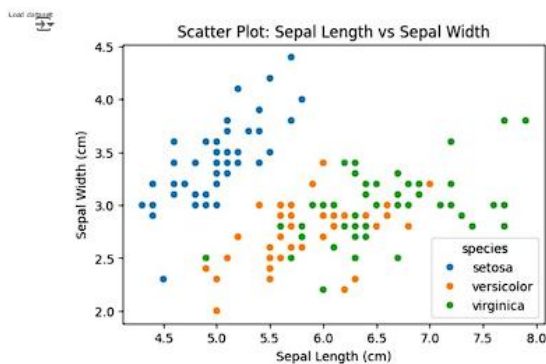
```
# Box Plot plt.figure(figsize=(6, 4))
sns.boxplot(x='species', y='petal_width', data=df)
plt.title('Box Plot: Petal Width per Species')
plt.xlabel('Species') plt.ylabel('Petal Width (cm)')
plt.show()
```



5. Scatter Plot

```
# Load dataset df = sns.load_dataset('iris')

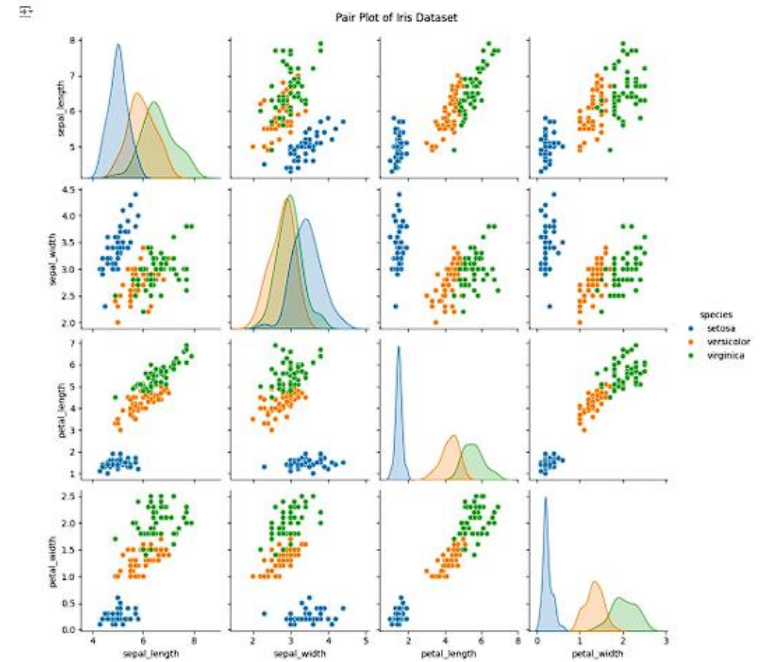
# Scatter Plot plt.figure(figsize=(6, 4)) sns.scatterplot(x='sepal_length',
y='sepal_width', hue='species', data=df) plt.title('Scatter Plot: Sepal Length
vs Sepal Width') plt.xlabel('Sepal Length (cm)') plt.ylabel('Sepal Width (cm)')
plt.show()
```



6. Pair Plot

```
# Load dataset df = sns.load_dataset('iris')
```

```
# Pair Plot sns.pairplot(df, hue='species')
plt.suptitle('Pair Plot of Iris Dataset', y=1.02) plt.show()
```



7. Heatmap (Correlation Matrix)

```
# Load dataset df = sns.load_dataset('iris')

# Heatmap plt.figure(figsize=(8, 6)) corr = df.drop('species',
axis=1)
Heatmap: Feature Correlation
fmt=
```

